Shaan Patel

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Education

University of Texas at Arlington

Ph.D. and M.S. in ***Physics and Applied Physics***, GPA: 4.00 — 2025

- Authored 6 peer-reviewed publications in leading astrophysics journals; presented research at 3 physics conferences

Georgia Institute of Technology

B.S. in ***Physics***, Highest Honor — 2019

Experience

Computational Astrophysics Researcher **| UT Arlington** — 2021 – 2025

- Executed 50,000+ numerical Python simulations with the Exoplanets group, optimizing mathematical algorithms to reduce computation time and improve efficiency by 80–90%
- Processed and visualized multi-million-row time-series datasets to identify key dynamical patterns

Physics Research Intern **| SLAC National Accelerator Lab** — 2019

- Worked with the Cryogenic Dark Matter Search (CDMS) team to test wiring and readout electronics for the He-3/He-4 dilution refrigerator

Lead Undergraduate Researcher **| Georgia Institute of Technology** — 2017 – 2019

- Led a research team in the Numerical Relativity Research Group simulating binary black hole dynamics on HPC clusters, analyzing high-dimensional gravitational wave data for LIGO research

Investment Analyst **| Georgia Institute of Technology** — 2016 – 2017

- Conducted research, financial modeling, and valuation analyses for the Student Investments Committee to manage and pitch equity investments for a multi-million-dollar student-run endowment

Projects

AuroraNet: Capstone Aurora Prediction Project

- Engineered an end-to-end time-series pipeline in Python (pandas, NumPy) to clean 24 years of space weather data, training PyTorch LSTM models to achieve an RMSE of 0.642
- Deployed real-time inference endpoints via FastAPI and built an interactive Streamlit dashboard, containerizing the app with Docker and logging experiments using Weights & Biases (W&B)

SolarWindLakehouse: End-to-End PySpark, Delta Lake & MLFlow Pipeline

- Built a Medallion Lakehouse anomaly detection engine on Databricks using PySpark and Delta Lake to process space telemetry, using PySpark SQL windowing for feature engineering and MLflow to track, register, and serve a SparkML K-Means pipeline for batch predictions

DefCoordML: NFL Play Prediction

- Developed ML models to predict NFL offensive plays (pass vs. rush) using historical game data
- Achieved 75% accuracy and 0.82 ROC AUC through feature engineering and model tuning

Skills

Languages & Systems: Python, SQL, Bash, Git, Linux, PyTest
Data Engineering & Cloud: Databricks, Apache Spark/PySpark, Delta Lake, Unity Catalog, Medallion Architecture
Machine Learning & MLOps: PyTorch, Scikit-learn, MLflow, Weights & Biases (W&B), Model Validation
Data Science & Analytics: A/B Testing, Hypothesis Testing, pandas, NumPy, SciPy, matplotlib, seaborn
Deployment: FastAPI, Streamlit, Docker
Generative AI & NLP: Retrieval-Augmented Generation (RAG), LLMs, Vector Databases, Semantic Search